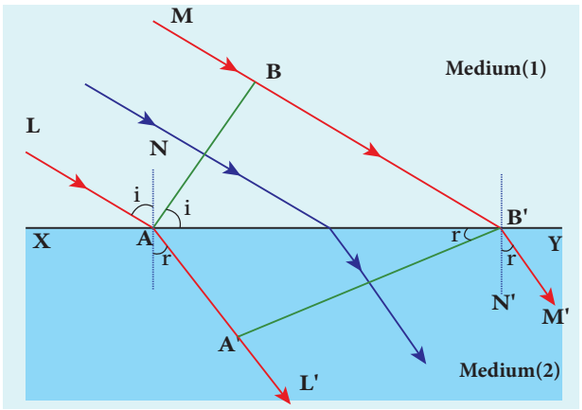


Figure 7.5 Law of refraction

to all the points on the wavefront. Thus, the refracted wavefront $A'B'$ emanates as a plane wavefront. The two normals N and N' are considered at the points where the rays L and M fall on the refracting surface. As refraction happens from rarer medium (1) to denser medium (2), the speed of light is v_1 and v_2 before and after refraction and v_1 is greater than v_2, $(v_1 > v_2)$. But, the time taken t for the ray to travel from B to B' is the same as the time taken for the ray to travel from A to A'.

$$t = \frac{BB'}{v_1} = \frac{AA'}{v_2} \text{ (or) } \frac{BB'}{AA'} = \frac{v_1}{v_2}$$

(i) The incident rays, the refracted rays and the normal are in the same plane.

(ii) Angle of incidence,

$i = \angle NAL = 90^\circ - \angle NAB = \angle BAB'$

Angle of refraction,

$r = \angle N'B'M' = 90^\circ - \angle N'B'A' = \angle A'B'A$

For the two right angle triangles $\Delta ABB'$ and $\Delta AA'B'$,

$$\frac{\sin i}{\sin r} = \frac{BB'/AB'}{AA'/AB'} = \frac{BB'}{AA'} = \frac{v_1}{v_2} = \frac{c/v_2}{c/v_1}$$

Here, c is speed of light in vacuum. The ratio c/v is a constant, called refractive index of the medium. The refractive index of medium (1) is, $c/v_1 = n_1$ and that of medium (2) is, $c/v_2 = n_2$.

In ratio form,

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \qquad (7.3)$$

In product form,

$$n_1 \sin i = n_2 \sin r \qquad (7.4)$$

Hence, the laws of refraction are proved.

In the same way the laws of refraction can be proved for wavefront travelling from denser to rarer medium also.

The speed of light is inversely proportional to the refrative index of the medium $(v \propto 1/n)$ and also directly proportional to wavelength of light $(v \propto \lambda)$. Hence,

$$\frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1} \qquad (7.5)$$

Note If light of a particular frequency travels through different media, then, its frequency remains unchanged in all the media. Only the wavelength changes according to speed of light in that medium.

EXAMPLE 7.1

The wavelength of light from sodium source in vacuum is 5893Å. What are its (a) wavelength, (b) speed and (c) frequency when this light travels in water which has a refractive index of 1.33.

Solution

The refractive index of vacuum, $n_1 = 1$

The wavelength in vacuum, $\lambda_1 = 5893$ Å.

The speed in vacuum, $c = v_1 = 3 \times 10^8$ m s^{-1}

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